PROCESS DIAGNOSTICS EVALUATION

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Date: 6/5/23 Process Name: Steering Link
Group: T Evaluator ID: Takt time:

Prod. Mgr Sign	White G/L	Red G/L	Total Score
SJ	JS	TR	30

Category	Process Design Evaluation Items	Definition	Judgement Criteria: ○	●	△	×	Before Kaizen: Judgement	Pt. Value	Sub Total	After Kaizen: Judgement	Sub Total
Safety	1 Ergonomic Evaluation	The physical difficulty of the process using a standard measurement method 1 Utilize NAMC shop specific measurement method 2 Target no high burden or "red" processes	• Low Burden (Green Process)		• Medium Burden (Yellow Process)	• High Burden (Red Process)	△	○ = 8 △ = 4 X = 0	4	△	4
Process Smoothness	1 Vehicle Sections	Count the # of vehicle sections that the TM works on during the process: 1 Vehicle section is defined as where the TM stands or sits 2 If TM goes from one section to another and then back to the original section then count as 3 sections 3 If TM works while walking then count as 1 section	• TM works in ≤ 2 vehicle sections	• TM must work in multiple vehicle sections for part completion	• TM works in n=3 vehicle sections	• TM works in ≥ 4 vehicle sections	○	○ = 3 ● = 3 △ = 2 X = 0	3	○	3
	2 Returns to Flowrack	Count the number of times the TM goes from the flowrack to the vehicle: 1 Use a shopping cart dolly to minimize the trips to the flowrack 2 Use "picnic" basket or pouch to minimize trips to the flowrack	• TM makes 0-1 returns to flowrack • TM taking a part from a dolly does not count as return		• n=2 Returns to the flowrack	• TM makes ≥ 3 returns to the flowrack	○	○ = 3 △ = 1 X = 0	3	○	3
	3 Prep Work	Confirm there is no prep work performed over WIP parts: 1 Prevent non-conformance related to dropped parts (No-op, UIN, mutilation, etc.)	• Established jig or prep table that is separated from WIP parts	• No Prep work being performed		• Prep work is being performed over WIP parts	●	○ = 1 ● = 1 X = 0	1	●	1
	4 Start / Stop Markers	Process Start and Stop Markers are Clear and easy to distinguish: 1 ▲ (Green) = Start 2 ▼ (Red) = Stop	• Markings are at appropriate locations • Markings are not damaged or peeling		• Markings are present and in appropriate location • Markings show signs of peeling / damage	• Markings are not present • Markings show significant damage or peeling	X	○ = 2 △ = 1 X = 0	0	○	2
	5 Hashmarks	Process hashmarks are clear and visible 1 Easy to distinguish if TM is ahead or behind 2 Good visual for TM to know where they are in their work pitch	• Present • Correct Number • Spacing (Axle x Axle Divide by 10	• Prep Job or Offline Job		• Markings are not present • Incorrect Number • Not evenly spaced	○	○ = 1 ● = 1 X = 0	1	○	1
	6 Work Time per Work Zone	Minimum amount of time required for a TM to be in a work zone 1 Introduce high risk of skipped elements if TM spends a small amount of time in a single work zone 2 E/G Room 20", FR Room 3", RR Room 17" is NG since FR Room is less than 5 seconds	• TM spends ≥5 seconds in a single work zone			• TM spends < 5 seconds in a single work zone	○	○ = 1 X = 0	1	○	1
	7 Pokayoke Judge Position	The pokayoke should trigger the Andon early enough to allow TL response time: 1 Design the process so the pokayoke dependent elements occur early in the process 2 TMMMS judgment zone is 70% of Pitch 3 High risk of frequent line stops when pokayoke element is performed late	• Pokayoke judgement zone ≤ 70% of pitch • Existing mark on floor designating judge position	• No pokayoke		• Judgement zone is > 70% • No marking present for judgement position	○	○ = 3 ● = 3 X = 0	3	X	0
	8 Process WACT	Calculate the weighted average cycle time for the process: 1 WACT = (Sum of Standard Work)+(Sum of option work x Option %)	• 90%-98% of Line Speed		<90% of Line Speed	• WACT > 98% of Line Speed	△	○ = 3 △ = 1 X = 0	1	○	3
	9 Process Fluctuation	Calculate the fluctuation of the process: 1 Fluctuation =[(CTmax-CT⚠n)/Line Speed]*100	<10%		• 11-20%	• > 20%	△	○ = 3 △ = 1 X = 0	1	○	3
	10 Process CTmax Ratio	Calculate the relationship of Max Cycle Time vs. Line Speed: 1 CTmax Ratio = [(Ctmax-Takt Time)/Takt Time]*100 2 TMMMS specifies acceptable standard	• Within +/- 5% of Line Speed			• Outside of +/- 5% of Line Speed	X	○ = 3 X = 0	0	○	3
	11 Value Added Work (VAW)	Time the value added work on the process 1 Walking, checking manifest, installing small component parts, etc is not value added 2 VAW = Set part to body, shoot bolts, hand start when necessary, clip part to body, route WH to final location, etc.	• ≥ 50%		• 35-50%	• ≤ 35%	○	○ = 4 △ = 2 X = 0	4	○	4
	12 Yamazumi	Tool used to visualize the individual work elements of a process 1 Facilitates work balancing 2 Visualizes Value vs. Non-Value added work 3 Reduces Muda	• Yamazumi matches PES/EIS • Element colors are correct White-Standard Yellow-Option Blue-Telescopic Gray-Overtime			• Yamazumi doesn't match EIS/PES • Element colors are incorrect	X	○ = 2 X = 0	0	○	2
	13 Parts Fit Within Work Zone	All parts installed should fit within TM work zone: 1 Eliminates Muda 2 Creates smoother process for TM 3 Work zone defined as when TM first touches the vehicle to when they leave the vehicle	• All parts fit within work zone of TM	• Only low option parts do not fit within work zone (<= 20% usage rate)		• Parts do not fit within work zone (TM has to leave work zone to retrieve part)	○	○ = 2 ● = 2 X = 0	2	○	2
	14 Process Interference	Processes should not interfere with one another 1 Interference can occur across (TM hands part to another TM 2 Can occur in consecutive L10's (confirm no dolly interference) 3 TMMMS Approved pair work is needed	• No assigned jobs in same zone in adjacent processes. No Interference with TM	• Approved Pair Work (ex. RR Seat Cushion)		• Interference Occurs	X	○ = 2 ● = 2 X = 0	0	○	2
	15 Work Space Secured	TM should be able to work in a safe environment 1 Interference with equipment / robot 2 Entrance / Exit to Line 3 Use critical eye for items related to process environment that pose a risk	• Available space > 1 pitch • Entrance / Exit to Line is clear and TM process does not interfere • No potential for slip, trip, fall • Range of machine motion acceptable			• Available space < 1 pitch • Entrance / Exit to Line not clear and/or TM process interferes • Potential for slip, trip, fall • Range of machine motion NG	○	○ = 2 X = 0	2	○	2

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	16 Tool / Jig Quantity	**Check to confirm number of tools TM has to use at process:** 1. TMs have to make decisions to pick correct tools or jigs to install parts, it adds complexity 2. Number of tools and jigs used in a process should be minimized (Less Stress, Simple)	• ≤ 2 tools / jigs per process • Have established "Home"		• 3-4 tools and/or jigs w/ established "Home"	• 5 or more tools and/or jigs • 2 tools that have the same size socket or bit • Tools/jigs do not have established "Home"	X	○ = 2 △ = 1 X = 0	0	X 0
	17 Tool / Jig Interference	**Check to confirm interference with tools in process:** 1. Tools should be free of interference from other tools, equipment, and/or parts	• No catching • No Sticking • No Dragging • No Tangles of Hoses • No Interference with Other Tools, equipment, parts			• Tool catches, sticks, or drags • Tools is tangled with others • Interference with Other Tools, equipment, parts	0	○ = 1 X = 0	1	0 1
	18 Impact Gun Type	**Check to confirm process has correct impact gun:** 1. Compare process to QCC	• Process matches QCC			• Process does not match QCC	0	○ = 1 X = 0	1	0 1
Flowrack Setup	1 Flowrack Parts Position (Including Parts Table)	**Parts should be located in the flowrack to minimize bending and reaching:** 1. Place high use parts at waist level and in the most convenient position 2. Place #9 parts (fasteners) on a separate flowrack if they're not picked every cycle by TM 3. Chuters and racks are located within 1 pitch 4. Fastener bins should not be located above WIP totes to prevent fasteners from being trapped in part and causing non-conformance	• High use parts are at waist level • Flowracks are located within the process pitch • Dedicated position for empties for isle racks between lines • Fastener bin is not located above WIP parts	• No flowrack at the process		• High use parts are not at waist level • Flowracks are not within 1 process pitch • Fastener bin located above WIP parts	●	○ = 1 ● = 1 X = 0	1	● 1
	2 Similar parts separate on Flowrack	**Separate parts of similar appearance to prevent a wrong part installation:** 1. Bolts, Screws, Nuts, Parts similar in shape/color, etc 2. Includes on dollies	• Similar parts are separated to prevent installation of wrong parts • Correct bin color used Kanban labeled and correct	• No similar parts on process		• Similar parts are left/right or up/down of each other and show risk of wrong part installation • Wrong Bin color used • Kanban is incorrect or bin is not labeled	●	○ = 2 ● = 2 X = 0	2	● 2
	3 Flowrack Labels	**Provide a clear visual to TM and Conveyance to ensure correct part installation and delivery** 1. Should be present on FR and RR of flowrack and each label should match 2. Kanban should be indicated with correct paper color and fastener be displayed	• Present Front/Rear • Kanban indicated • Correct paper color • Fastener displayed front & rear			• Not Present Front/Rear • No kanban indicator • Incorrect paper color • No fastener displayed	0	○ = 1 X = 0	1	0 1
	4 Dunnage Handling	**Handling of boxes/dunnage should be minimized to create a smooth process for TM:** 1. Handling of packaging materials inside a box 2. Changing out empty part boxes 3. Changing out dollies (ex. Door Trim Dollies)	• Frequency ≥ Every 10 Vehicles	• No dunnage/box handling at process	• Frequency ≤ Every 10 Vehicles and time allotted in EIS	• Frequency æ Every 10 Vehicles and NO time allotted in EIS	●	○ = 2 ● = 2 △ = 1 X = 0	2	● 2
	5 Flowrack / Chuter Condition	**Ensure Flowrack and Chuters are in good operating condition** 1. Poor condition could result in TM spending excess time getting a tote to the proper location	• Sufficient space for part boxes and empties • No sticking of totes or trays	• No chuter or flowrack present		• Totes / Boxes / Trays get stuck and require extra tool or time to remove stuck condition	●	○ = 1 ● = 1 X = 0	1	● 1
Process w/o SPS	1 Parts Location	**Confirm parts are located in same order as the job sequence:** 1. Creates smoother process 2. Helps eliminate extra returns 3. Manifest symbols are fresh on TMs mind	• TM picks part per sequence of manifest symbols			• TM does not pick part per sequence of manifest symbols	X	○ = 1 X = 0	0	0 1
	2 Parts Handling	**Confirm TM is not transferring part from LH to RH or RH to LH:** 1. Muda for TM if they are switching hands and do not need to. Creates extra step and does not support efficient process	• TM picks part from box and installs with the same hand used to pick part (no transfer)			• TM transfer part from RH to LH or LH to RH after picking	X	○ = 1 X = 0	0	0 1
	3 Pick Light	**Confirm Pick Light is Installed for Multiple spec parts:** 1. Pick light helps TM identify which part to install on vehicle 2. Should be utilized in processes that have multiple specs for a part	• Pick light present if ≥ 2 variations of a part • Light works correctly • No damage to light	• Only n=1 variation of a part so pick light is not required		• Pick light not present if ≥ 2 variations of a part • Light does not work • Damage to light	0	○ = 1 ● = 1 X = 0	1	0 1
Process W/ SPS Box	1 Parts Placement	**Confirm parts are placed in the order of Assy Sequence** 1. Improves overall flow of process when TMs need to utilize SPS box	• Placed in order of Assy Seq	• No SPS Box		• Not placed in order of Assy seq	X	○ = 1 ● = 1 X = 0	0	0 1
	2 Parts Interference	**Confirm no interference with other parts in SPS box** 1. Reduce risk of mutilation 2. Reduce risk of part entanglement	• Parts do not contact other parts in SPS box	• No SPS Box		• Parts contact other parts in SPS Box	X	○ = 1 ● = 1 X = 0	0	0 1
	3 Parts Picking	**Confirm TM is picking from 1 SPS box in their process:** 1. Reduce risk of missed step 2. Smoother process, less muda	• 1 Process / 1 SPS Box TM installs part picked in same zone	• No SPS Box	• 1 Process / 2 SPS Box ONLY if part from second box is installed in same TM work zone	• 1 Process / 2 or more SPS box and not installed in same work zone	0	○ = 2 ● = 2 △ = 1 X = 0	2	0 2
	4 SPS Box Handling	**Confirm TM Does not Move SPS Box when Picking:** 1. Box should be set in 1 of 4 vehicle zones 2. TM should not have to move box to pick part and then put box back	• SPS box stays in same location when part is getting picked. Only time SPS box should be moved is if it is being returned	• No SPS Box		• TM moves SPS box to pick part	0	○ = 1 ● = 1 X = 0	1	0 1
	5 Parts Handling	**Confirm TM is not transferring part from LH to RH or RH to LH:** 1. Muda for TM if they are switching hands and do not need to. Creates extra step and does not support efficient process	• TM picks part from box and installs with the same hand used to pick part (no transfer)	• No SPS Box		• TM transfer part from RH to LH or LH to RH after picking	X	○ = 1 ● = 1 X = 0	0	0 1
Documentation	1 EIS Instruction	**Check EIS for key information for TM to perform process successfully:** 1. Contain all necessary key information (How to, Good vs. NG, ID parts, etc.)	• EIS contains clear, concise "How To" instructions for all parts, including how to hold • Clear or Measurable judgment criteria for all parts installed or inspected			• EIS does not contain clear, concise "How To" instructions for all parts • Does not contain clear or Measurable judgement criteria for all parts installed or inspected	X	○ = 2 X = 0	0	0 1
	2 EIS Non-Compliance	**Check EIS for effects of non-compliance comments:** 1. Allows TM to understand potential defect caused by not completing the element performed	• EIS contains effects of non-compliance as per shop standard			• EIS does not contain effects of non-compliance as per shop standard	X	○ = 2 X = 0	0	0 1
	1 Standard Dolly	**Check if dolly in process is a standard dolly:** 1. Should not be able to place empty boxes on dolly 2. Has temp home for parts that are picked	• TM cannot place empty box on dolly • Has temp home for picked parts and home is large enough to accommodate part • Mutilation coverings protecting vehicle	• Dolly not on Process	• TM cannot place empty box on dolly • No temp home for picked parts OR No Mutilation coverings protecting vehicle	• TM can place empty box on dolly • No Temp Home for Picked Parts • Dolly is damaged/dirty	0	○ = 2 ● = 2 △ = 1 X = 0	2	0 2

Dollies	2 Nuts & Bolts	**Confirm dolly has location for nuts and bolts:** 1 Should be < 2 hours worth 2 Fastener tote needs Max line 3 Visual controls necessary for TM to refill or call for help 4 No WIP parts stored below totes-prevent nuts, bolts, screws from getting trapped	• ≤ 80% of fastener tote capacity • Capacity line is clear • No WIP parts stored below fastener tote	• No Fasteners on process or no totes on dolly		• > 80% of tote capacity Capacity line is not clear • Fastener tote stored above WIP parts	0	○ = 1 ● = 1 X = 0	1	0 1
	3 Part Replenishment	**Check if part replenishment lamp is installed on dolly:** 1 Visual and/or Andon control for TM to alert TL that parts replenishment is needed 2 Should be utilized to prevent disruption to process(es)	• Lamp present, functions, and easy to see OR • Andon Present and easily accessible	• Process has no parts to replenish		• No lamp present and process has parts to replenish • Lamp does not function • Andon not Present	0	○ = 1 ● = 1 X = 0	1	0 1
	4 Operation	**Confirm dolly operation:** 1. TM should be able to operate without issue to allow for smooth, efficient, process	• Dolly should move smoothly and not stick or have any interference	• No dolly at process		• Dolly is difficult to operate • Dolly gets stuck or has other interference	X	○ = 1 ● = 1 X = 0	0	0 1
	5 Chameleon / Vacuum	**Check that vacuum and/or chameleon functions properly:** 1 NG function may result in TM overburden 2 Chameleon are magnets that pick up desired number of fasteners	• Vacuum and Chameleon function correctly • Chameleon picks correct number of fasteners • Speed of chameleon does not allow for dropped parts	• No fasteners on dolly	• Chameleon picks up fasteners, but not the correct number	• Vacuum does not function • Chameleon does not pick up fasteners or fasteners are dropped		○ = 2 ● = 2 △ = 1 X = 0	2	2
	6 Scrap Parts	**Confirm dollies have designated location for scrap parts:** 1 Fasteners only Main parts are excluded	• Location and size of scrap bin adequate • No overflowing bins (Max capacity is clear) • Bins are secured to dolly	• No fasteners on dolly		• Location NG • Size NG • Overflowing Bin • Max capacity line not clear • Bins not secured	0	○ = 1 ● = 1 X = 0	1	0 1
	7 Ergonomics	**Confirm TM does not have to turn excessively to pick parts, tools, or jigs:** 1 Reduce TM burden 2 Eliminate muda 3 Parts, tools, jigs should be easily reached by TM	• ≤ 90 degree turn to pick part, tool, or jig	• No dolly at process		• > 90 degree turn to pick part, tool, or jig	0	○ = 1 ● = 1 X = 0	1	0 1
Quality	1 Part Completion	**Check to see if the major parts are completely installed on one process:** 1 Use filled circle rating if the part cannot be completely installed due to vehicle structure, ergonomic reasons (LH/RH CSA), or TAKT Time 2 Grommets do not have to be installed with the major part, only judge the installation of the part (shoot, connect, etc)	• Part completely installed on 1 process	• Cannot do due to vehicle structure, ergonomics, or takt time		• NG not done but possible - no structural ergonomic, or TAKT time issues preventing part completion	0	○ = 6 ● = 6 X = 0	6	0 6
	2 QA Network Required for all ▽▽▽Ⓓ items	**Used to ensure proper controls are in place to prevent critical flowout:** 1 Identify ▽ and / or FUNC A critical items for flow out risk evaluation 2 Determine C/M to reduce flow out for ▽ and / or FUNC A critical items 3 Each process to have a QAN score to determine necessary controls needed to prevent critical flowout	• QCPS assigned to correct group • Proper Signage at process board • Primary tool correct & documented • Back up tool correct & documented •Repair log contains QCPS # & Name Group # and Process Name Correct picture / Joint #'s • QCPS Name & Number listed Pokayoke Box • Process book contains QAN Score	• No ▽ ▽▽Ⓓ items on process		• Process does not meet all of the required criteria for ▽ ▽▽Ⓓ • Process book does not contain QAN score	0	○ = 6 ● = 6 X = 0	6	0 6
	3 Manifest Matches Parts I D	**Flowrack Label and part ID matches the Manifest Symbol:** 1. Does the manifest symbol match the flowrack label? 2 Does the part have an ID Mark? 3 If #2 is YES, does the flowrack label match the part ID?	• Manifest symbol, parts ID & flowrack label all match	• Manifest not required		• Manifest symbol, parts ID or flowrack label do not match		○ = 2 ● = 2 X = 0	2	2
	4 # of Manifest Blocks/Layout	**Count the number of manifest blocks the TM reads on the process:** 1 Blocks should also be on the same line 2 Do not count general information at the top of the manifest (seq, body, etc)	• ≤ 3 blocks on the same line	• No manifest blocks used	• 4 blocks and on the same line	• ≥ 5 blocks or blocks are not on the same line		○ = 2 ● = 2 △ = 1 X = 0	2	2
	5 Rare Parts / Option Parts	**Check to see if process has system to alert TM:** 1 Reduce potential for wrong or missed part 2 Rare parts can disrupt TM rhythm 3 Any part installed on less than 7% is classified as rare (ex Clutch) 4 Option parts should be picked first	• Pokayoke or warning system installed to alert TM of RARE part • (5 parts or less per shift) • Option parts are picked first in process	• Process doesn't have any rare parts or option parts	• Rare/Option parts are sequenced with other parts	• TM read manifest only • Option parts are not picked first		○ = 2 ● = 2 △ = 1 X = 0	2	2
	6 Mutilations	**Check to see if system damage potential exists:** 1 Use TMMMS shop specific standards to evaluate this item 2 Includes dollies, flowracks, tools, SPS Box, My cover condition	• Dollies/Equipment padded Gun covers/Sleeves utilized & in good condition • No contact with painted surfaces • My cover in good condition	• Mutilation Diagnostics not required No painted surfaces in work area	• TM or Part contacts painted surface with Mutilation Cover utilized	• Contact with painted surface w/o mutilation cover • Dollies / Tools / Equipment not covered • My cover in NG condition		○ = 6 ● = 6 △ = 3 X = 0	6	6
Max/Min Ratio = $\frac{Cmax - Cmin}{Takt\ Time} \times 100$		CT Max Ratio = $\frac{CT\ max - Takt\ Time}{Takt\ Time} \times 100$	Corolla Max CT		*When Possible Perform Cycle Time on n=5 Standard and n=5 with Option to Understand Max/Min CT		Before Score	0		After Score
								30		31